Design Evaluation 3 and Reflection

XR Study Whiteboard

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Objective and Validation Metrics

The final IP2b test examined whether students could follow a board lesson and take notes in VR. It retested the seated view, paper tools, drawing, navigation, tracking, and replay after the earlier usability problems. This evaluation uses the final five-person headset test, documented in the Iteration 2 records. Success measures came from the existing testing plan. The aim was to judge whether the classroom workflow was usable before considering wider educational benefits.

Table 1

Planned Success Measures

Aim	Criterion
Shared view and note taking	At least four of five view both surfaces and copy the lesson content.
Paper tools and drawing	At least four of five use the tools. Discovery within 15 seconds and 90% correct strokes were also planned.
Navigation and replay	At least four of five move and complete Record, Stop, and Play. Controllers remain aligned.
Confidence and comfort	Median confidence is at least 4/5 with no severe comfort, tracking, or data loss issue.

Method and Results

Five classmates used the Unity build, a headset, and two physical controllers in moderated task tests. Kai had a voice recording and Syifa had a camera recording. Amy, Serafina, and Safira have notes only. All five completed the eight recorded task categories, including copying the date, line, and circle onto paper. All six rating measures were 5/5 for every participant. No severe comfort problem or technical blocker was recorded. Appendix A contains the results and Appendix B contains the supplied session records.

Analysis and Insights

Table 2

Post-Test Ratings Across Five Participants

Measure	P01	P02	P03	P04	P05	Median
Seated view comfort	5	5	5	5	5	5
Paper drawing reliability	5	5	5	5	5	5
Paper control clarity	5	5	5	5	5	5
Movement predictability	5	5	5	5	5	5
Recording control clarity	5	5	5	5	5	5
Overall confidence	5	5	5	5	5	5

The shared view supported the study task. Kai described looking between the board and paper without moving away from the desk. All five completed note copying. Comfortable placement therefore deserves priority over adding classroom decoration. The fixed paper also gave the right trigger a clear purpose. These results are consistent with the earlier viewpoint and paper fixes, although the study did not isolate each change.

Task completion did not mean immediate understanding. Kai initially explored the teleport controls before understanding them. Syifa reopened the guide and received instructions about colour selection and recording order. The transcript includes a second recording exercise, despite the checklist listing one attempt. I therefore report completion with support rather than five independent first-attempt successes. The guide helped recovery, but independent discovery remains uncertain.

All five completed replay, and Kai expected it to reproduce his marks. Syifa said playback matched her expectation. This supports replay as a review tool. It does not prove that replay improves recall. Likewise, positive remote-learning comments show interest, not increased concentration. The highest possible score on every rating also hides small differences between users and should be read alongside the observed need for help.

Evaluation of Aims

The shared view and lesson-to-notes aims were met in all five sessions. Navigation and physical controller tracking also passed across the group. Recording was completed by everyone, with explicit moderator support in Syifa's transcript. These findings support the tested workflow under moderated conditions. No severe comfort problem was recorded, but short sessions cannot establish comfort during a full lesson.

Paper tools and drawing were supported by task completion and ratings. The planned 15-second discovery target cannot be verified because selection times and prompt levels were not logged. The 90% stroke target cannot be calculated without raw stroke and defect counts. Those two thresholds remain uncertain. Simulator T/Y controls and hand tracking were outside the physical-controller test. No aim was disproved, but unmeasured aims should not be labelled as passed.

Updated Concept and Design Decisions

XR Study Whiteboard is a VR classroom where a learner follows the shared board while writing on fixed personal paper. The left controller handles navigation and the right controller selects tools and writes. Visible tool feedback and board-action replay support the study activity. The concept is partially confirmed. Participants could use and understand it, but better learning and sustained attention remain untested.

I would keep the seated viewpoints, fixed paper, and clear controller roles. I would improve first use with a brief practice task and a guide that remains easy to reopen. Recording needs clearer Ready, Recording, Stopped, and Playing states because Syifa needed sequence reminders. Saved notes should be tested before larger rooms or multiplayer, since preserving study work directly supports the core task. These are proposed refinements, not features validated by this test.

Final Reflection

Prototype Session Review

I chose Syifa's final IP2b session. She copied board content and completed paper tools and replay. Keeping both surfaces visible supported note taking. However, she reopened the guide and needed recording instructions. Her 5/5 ratings showed satisfaction, while the transcript revealed support needs that the pass checklist missed.

Methodology and Concept

Paper Wizard-of-Oz testing helped me explore gesture meanings quickly, but manually simulating responses could not test tracking. IP1 exposed hidden paper tools and viewing problems. The IP2a/IP2b development cycle moved this work into headset testing. Think aloud revealed expectations, but my leading questions and demonstrations reduced evidence of independent use. Consistent tasks helped comparison, yet three notes-only records and missing timings weakened measurement. I learned to separate task completion from learning effectiveness. The classroom interaction is supported, while the wider educational claim is only partially confirmed.

Improvements and Extensions

I would test five new users with neutral prompts and log time, assistance, stroke defects, and comfort. A longer lesson should check note quality and later recall. I would also prototype passthrough AR with a virtual board above a real desk. This might support physical note taking, but anchoring, readability, and distractions need comparison with VR before choosing it. AR might also weaken the classroom feeling that motivated VR.

Appendix A

Participant Results and Evidence Register

The facilitator confirmed that the five-person Iteration 2 headset dataset represents the final IP2b/Week 12 test. Participant IDs here belong to that dataset. They should not be combined with the earlier IP1 IDs as if they identify the same people. Only Kai and Syifa have recording links. The other three records are observation notes, not reconstructed interviews.

Table A1

Evidence Available for Each Participant

ID	Name	Evidence
P01	Kai	Voice recording and supplied transcript
P02	Syifa	Camera recording and supplied transcript
P03	Amy	In-class notes only
P04	Serafina	In-class notes only
P05	Safira	In-class notes only

Table A2

Observed Task Completion

Task	P01	P02	P03	P04	P05
Explore	Pass	Pass	Pass	Pass	Pass
Seat and shared view	Pass	Pass	Pass	Pass	Pass
Follow and take notes	Pass	Pass	Pass	Pass	Pass
Paper tools and strokes	Pass	Pass	Pass	Pass	Pass
Move and return	Pass	Pass	Pass	Pass	Pass
Whiteboard workflow	Pass	Pass	Pass	Pass	Pass
Record and replay	Pass	Pass	Pass	Pass	Pass
Physical controller tracking	Pass	Pass	Pass	Pass	Pass
Simulator T/Y selection	N/A	N/A	N/A	N/A	N/A

The source checklist lists one attempt per category. Syifa's supplied transcript contains a repeated recording exercise and several prompts. Pass therefore means the task was completed during a moderated session. It does not establish that every task was completed independently on the first attempt. Table 2 in the assessed response reports all six supplied ratings for each participant. Informal 10/10 satisfaction comments are separate from those 1-to-5 measures.

Recording Evidence

Kai voice recording

Syifa camera recording

The supplied transcripts and notes are the basis of this report. The linked media have not been independently retranscribed for this submission. Kai's transcript refers to mouse input in one answer. The facilitator's note identifies the session as headset and physical-controller testing. Both statements are retained in Appendix B so the discrepancy remains visible.

Appendix B

Supplied Interview Transcripts and Notes

P01: Kai

Nayaka: Good afternoon. My name is Nayaka, and I am going to guide you through testing my XR Study Whiteboard. Before we start, may I know your name?

Kai: Yeah, my name is Kai.

Nayaka: Nice to meet you.

Kai: Nice to meet you too.

Nayaka: Before we start, you are using the Unity VR build with the headset and two physical controllers. Can you see both controllers in the environment?

Kai: Yeah, I can see both of them and they are following my hands.

Nayaka: First, can you tell me what you see in the XR environment?

Kai: It looks like I am in a classroom right now. I can see a whiteboard. It says XR Study Whiteboard and Study Session. I can also see black, red, marker, eraser, Ready to Record, and Play.

Nayaka: Can you see all the buttons?

Kai: Yeah. Whiteboard, Stand, Jump, Center Table.

Nayaka: Can you try pressing one of them?

Kai: Yeah.

Nayaka: What do you think these buttons do? What were you expecting?

Kai: Initially, I did not know what to do. But now that I have pressed it, I can see that it teleports me to one of the tables depending on which one I press. If I press Stand, it puts me back.

Nayaka: When you move, do both VR controllers stay in the correct position with your hands?

Kai: Yeah, both of them stay with me.

Nayaka: Okay. Can you move to the whiteboard?

Kai: Yeah.

Nayaka: After teleporting to the whiteboard, what do you think you can do here?

Kai: Looking at the marker and everything, I expect that I could act like a teacher and write something on the whiteboard.

Nayaka: Okay. Can you try it?

Kai: Hello.

Nayaka: What do you think the eraser does?

Kai: I imagine it erases whatever I just wrote.

Nayaka: Nice. What other controls can you see?

Kai: Clear Board.

Nayaka: What do you expect that to do?

Kai: Clear my whole board.

Nayaka: Yes, that is right. What about changing colours?

Kai: Yeah, I can change the colour.

Nayaka: Do you see any other buttons?

Kai: There is a Record button.

Nayaka: What do you think that does?

Kai: It says Ready to Record, and that it records board actions and timings. I am assuming it records what I write. Then when I press Play, it will repeat what I wrote.

Nayaka: Can you try it?

Kai: For example, if I write “Hello”, then press Stop and Play, I would imagine it clears the board first and then writes “Hello” again.

Nayaka: Yes, that is right.

Kai: Oh, okay.

Nayaka: Great. We have finished testing the whiteboard. Can you now try the features on the desk?

Kai: Yep. Now I am at one of the desks. I am pressing the left click button on the paper to write “Hello”. I am going to press Eraser, and I expect it to erase what I wrote.

Nayaka: Yes.

Kai: Now I am going to write a lot on the paper. Then I will press Clear Paper. I expect it to clear the paper.

Nayaka: Okay. While you are at the desk, can you see the paper and the teaching board comfortably from the seated position?

Kai: Yeah. I can look between the paper and the board without needing to move away from the desk.

Nayaka: Can you copy the date, horizontal line, and circle from the teaching board onto the paper?

Kai: Yeah.

Nayaka: How did that feel?

Kai: Pretty easy. I can look at the board and then write it on the paper.

Nayaka: Okay. What do you think the overall purpose of this prototype is, and who do you think would use it?

Kai: I think it is like a digital classroom. It could be for students or teachers. Maybe it would be useful if something similar to COVID happened again and people had to learn from home.

Nayaka: Yes. One of the ideas behind this is that during COVID, students could sometimes find it difficult to stay focused when classes were only through something like Zoom. With this, students could feel more like they are actually inside a classroom.

Kai: Yeah.

Nayaka: From one to five, how difficult do you think the prototype is to use?

Kai: It is not very difficult because I am using a mouse. If I was using a VR controller, I do not think it would be much harder either. It is pretty easy overall.

Moderator note: The session itself was completed using the VR headset and physical controllers. The wording above is retained from the original recording transcript.

Nayaka: Thinking specifically about the VR controllers you are using now, did they feel easy to control?

Kai: Yeah. They are easy to use and both controllers respond properly.

Nayaka: Would you use something like this if you were in a situation where you needed to attend classes remotely?

Kai: Yeah. The idea is cool, especially because you are inside a classroom instead of being alone at home. I think I would definitely use it.

Nayaka: Thank you. Do you have any feedback or anything you think I should improve?

Kai: It seems pretty good. I think everything is pretty clear. I am just trying to think of something. The writing and deleting controls are clear. I do not really have anything else to add. It is pretty good.

Nayaka: Okay. Thank you so much for your time.

Kai: No worries.

P02: Syifa

Nayaka: Okay, I am going to do VR testing. May I know your name?

Syifa: Syifa.

Nayaka: Can I record this interaction?

Syifa: Yeah.

Nayaka: Okay. First, can you grab the two controllers? What do you see on the screen?

Syifa: There is a pop-up guide.

Nayaka: Before we continue, can you see both physical controllers inside the VR environment?

Syifa: Yeah.

Nayaka: Are both controllers following your hands correctly?

Syifa: Yeah.

Nayaka: What does it say?

Syifa: Controls.

Nayaka: Okay. Press A. The A button is on your right controller. What can you understand from the guide?

Syifa: There are instructions for the controls.

Nayaka: So you understand the guide?

Syifa: Yeah.

Nayaka: Can you move from one table to another table?

Syifa: Like this?

Nayaka: Yes. Maybe close the guide first and then try moving to another table.

Syifa: Okay.

Nayaka: Did both physical controllers stay aligned with your hands when you moved?

Syifa: Yeah.

Nayaka: Can you look around? What do you see on the table?

Syifa: Paper.

Nayaka: What else?

Syifa: The instructions.

Nayaka: Can you also see the teaching board from where you are sitting?

Syifa: Yeah.

Nayaka: Can you look between the teaching board and the paper comfortably?

Syifa: Yeah.

Nayaka: Can you copy the date, horizontal line, and circle from the board onto your paper?

Syifa: Okay.

Nayaka: Was that comfortable?

Syifa: Yeah.

Nayaka: Can you try writing something on the paper? You can look at the guide again if you need help.

Syifa: Okay.

Nayaka: Now try changing the tool to the eraser.

Syifa: Okay.

Nayaka: Can you erase what you wrote?

Syifa: Yeah.

Nayaka: What other button can you see?

Syifa: Clear.

Nayaka: Can you try clearing the paper?

Syifa: Okay.

Nayaka: Now can you move to the whiteboard?

Syifa: Okay.

Nayaka: Nice. I can see you are looking at the guide again. You can close it when you are ready. What do you see on the whiteboard?

Syifa: The tools and colours.

Nayaka: You can move around using both controllers. Try moving forward. Now can you write something on the whiteboard using the red marker? You can check the guide again if you are confused.

Syifa: Okay.

Nayaka: Nice. What else can you see on the whiteboard interface?

Syifa: The colours.

Nayaka: What else?

Syifa: The tools. Eraser.

Nayaka: Anything else?

Syifa: Record.

Nayaka: Can you try the Record button? Before that, can you clear the whiteboard?

Syifa: Okay.

Nayaka: Now press Record and write something on the whiteboard. Maybe use the blue marker. Select Marker first, then select Blue.

Syifa: Okay.

Nayaka: Nice. Now press Stop and then try the Play button.

Syifa: Okay.

Nayaka: Let us try it again. Draw something, but make sure you press Record first. After you finish, press Stop, and then press Play.

Syifa: Okay.

Nayaka: What do you understand from this activity?

Syifa: We can write something on the whiteboard, draw something on the paper, and record it as well.

Nayaka: Did the replay match what you expected?

Syifa: Yeah.

Nayaka: And while moving and drawing, did the left and right VR controllers stay responsive?

Syifa: Yeah, both of them worked properly.

Nayaka: Yes. The purpose is to create a virtual classroom so that students who are studying remotely, such as through Zoom, can feel more like they are inside a real classroom and hopefully pay more attention to the teacher. Would you use something like this if you were studying remotely, like during COVID?

Syifa: Yeah.

Nayaka: How would you rate the interaction activities?

Syifa: Ten.

Nayaka: Ten out of ten?

Syifa: Yeah.

Nayaka: Do you have any feedback or anything you think should be improved?

Syifa: No, it is already good.

Nayaka: Okay. Thank you for your time.

P03: Amy (notes-only session)

P03 was tested in class with a VR headset and two physical controllers. No audio or video recording was made.

Table B1

Supplied Participant Record

Task	Result	Attempts	Observation
Explore	PASS	1	Recognised the virtual classroom and study purpose.
Seat and shared view	PASS	1	Comfortably viewed the teaching board and paper.
Follow and take notes	PASS	1	Copied the required board content.
Paper tools and strokes	PASS	1	Used Pencil, Eraser, and Clear Paper.
Move and return	PASS	1	Navigation was stable and predictable.
Whiteboard workflow	PASS	1	Completed the required drawing activities.
Record and replay	PASS	1	Completed Record, Stop, and Play.
Physical controller tracking	PASS	1	Both controllers tracked correctly.

All six post-test measures were recorded as **5/5**. Overall satisfaction was **10/10**. remote-learning interest was **Yes**. No major issue was recorded. Optional additional classroom features were suggested as future work, with no core usability change requested.

P04: Serafina (notes-only session)

P04 was tested in class with a VR headset and two physical controllers. No audio or video recording was made.

Table B2

Supplied Participant Record

Task	Result	Attempts	Observation
Explore	PASS	1	Understood the classroom and study purpose.
Seat and shared view	PASS	1	Board and paper were visible and comfortable.
Follow and take notes	PASS	1	Completed the note-copying activity.
Paper tools and strokes	PASS	1	Drew, erased, and cleared successfully.
Move and return	PASS	1	Movement was predictable with stable tracking.
Whiteboard workflow	PASS	1	Completed all required whiteboard activities.
Record and replay	PASS	1	Recording and playback worked successfully.
Physical controller tracking	PASS	1	Both controllers remained correctly aligned.

All six post-test measures were recorded as **5/5**. Overall satisfaction was **10/10**. remote-learning interest was **Yes**. No major issue was recorded. A short introduction for first-time VR users was noted as a useful refinement.

P05: Safira (notes-only session)

P05 was tested in class with a VR headset and two physical controllers. No audio or video recording was made.

Table B3

Supplied Participant Record

Task	Result	Attempts	Observation
Explore	PASS	1	Understood the classroom and study concept.
Seat and shared view	PASS	1	Saw the paper and board comfortably.
Follow and take notes	PASS	1	Completed the board-to-paper activity.
Paper tools and strokes	PASS	1	Drawing and paper controls worked correctly.
Move and return	PASS	1	Navigation was stable and predictable.
Whiteboard workflow	PASS	1	Completed the whiteboard activity.
Record and replay	PASS	1	Completed Record-Stop-Play.
Physical controller tracking	PASS	1	Both controllers tracked correctly.

All six post-test measures were recorded as **5/5**. Overall satisfaction was **10/10**. remote-learning interest was **Yes**. No major issue was recorded. Saved notes or additional classroom environments were recorded as optional future features.

Appendix C

Evidence Audit and Development Trace

Earlier Prototype Evidence

Two participants tested gesture meanings in the Week 2 paper prototype. Simulated responses could not establish tracking quality. IP1 has three recorded interviews and two checklist-only sessions. It identified hidden paper tools, one paper-drawing complaint, inconsistent teleport discovery, and viewing difficulty. Later headset results suggest improvement, but different participants and input conditions prevent a controlled comparison.

Implementation Trace

Repository commits document seat and simulator fixes in 76d76ece, drawing and replay in 279ef661, study controls in 1661a5b4, Quest navigation in 11375c85, and paper interaction in 88a59400. These records show development. The participant records provide the usability evidence.

Audit of Claims

The records support task completion, ratings, reported controller alignment, and short-session comfort. Discovery time, stroke percentages, prompt counts, learning gain, long-session comfort, and independent first use remain unmeasured. AR, saved notes, and new studies are proposals. No new participant data have been added.

Source Record

The repository preserves the complete final session records, IP1 evidence, and paper prototype record. Participant findings and development history are kept distinct throughout this evaluation.